Fractions Made Fun: Dividing Fractions

Let's learn how to divide fractions step-by-step! [cite: 1, 3]

What are Fractions?
A fraction is a part of a whole! When we divide something into equal parts, each part is a fraction. For example, if you cut a pizza into 4 equal slices and eat 1, you ate $1/4$ of the pizza! [cite: 6, 7]

The Essentials of Division

Division means splitting into equal parts or sharing equally among groups. For example, $6 \div 2 = 3$. If you have 6 cookies and share them with 2 friends, each gets 3! [cite: 10, 11, 12]

Quick Review [cite: 18, 19]

- **Numerator:** The top number (how many parts we have) [cite: 20]
- **Denominator:** The bottom number (total equal parts) [cite: 20]
- **Reciprocal:** The "flipped" version of a fraction (e.g., $1/2$ becomes $2/1$) [cite: 38, 39]

The 5-Step Method: Keep, Change, Flip!

Dividing fractions means finding out how many times one fraction fits into another [cite: 24]. Follow these steps:

Step 1: KEEP

Keep the first fraction exactly as it is [cite: 27, 28].

Step 2: CHANGE

Change the division sign (÷) to a multiplication sign (×) [cite: 30, 31, 32].

Step 3: FLIP

Flip the second fraction (find the reciprocal) [cite: 36, 37, 38].

Step 4: MULTIPLY

Multiply straight across: numerator × numerator and denominator × denominator [cite: 41, 43, 44, 45].

Step 5: SIMPLIFY

Simplify the result to the lowest terms or a mixed number if needed [cite: 49, 50, 51].

Walkthrough Example: $2/3 \div 4/5$ [cite: 57]

- Step 1: Keep $2/3$ [cite: 58]
- Step 2: Change to × [cite: 59]
- Step 3: Flip $4/5$ to $5/4$ [cite: 60]
- Step 4: Multiply $2/3 \times 5/4 = 10/12$ [cite: 61]
- Step 5: Simplify $10/12 = 5/6$ [cite: 62]
- **Final Answer:** $5/6$ [cite: 63]

Real-Life Practice

1. **Chocolate Bars:** Priya has ***3/4*** of a chocolate bar and shares it among 3 friends. How much does each get? [cite: 66, 67]
2. **Recipe Ingredients:** Arjun needs ***2/3*** cup of flour. His measuring cup holds ***1/6*** cup. How many scoops does he need? [cite: 68, 69]
3. **Ribbon Pieces:** Meera has ***4/5*** meter of ribbon and cuts it into ***1/10*** meter pieces. How many pieces can she make? [cite: 70, 71]

Watch Out! Common Mistakes [cite: 72, 73]

- Forgetting to flip the second fraction [cite: 74].
- Multiplying before flipping [cite: 74].
- Forgetting to simplify the final answer [cite: 74].

SUMMARY

Keep, Change, Flip, Multiply, and Simplify! [cite: 92]

Math is fun when we learn step-by-step! [cite: 91]